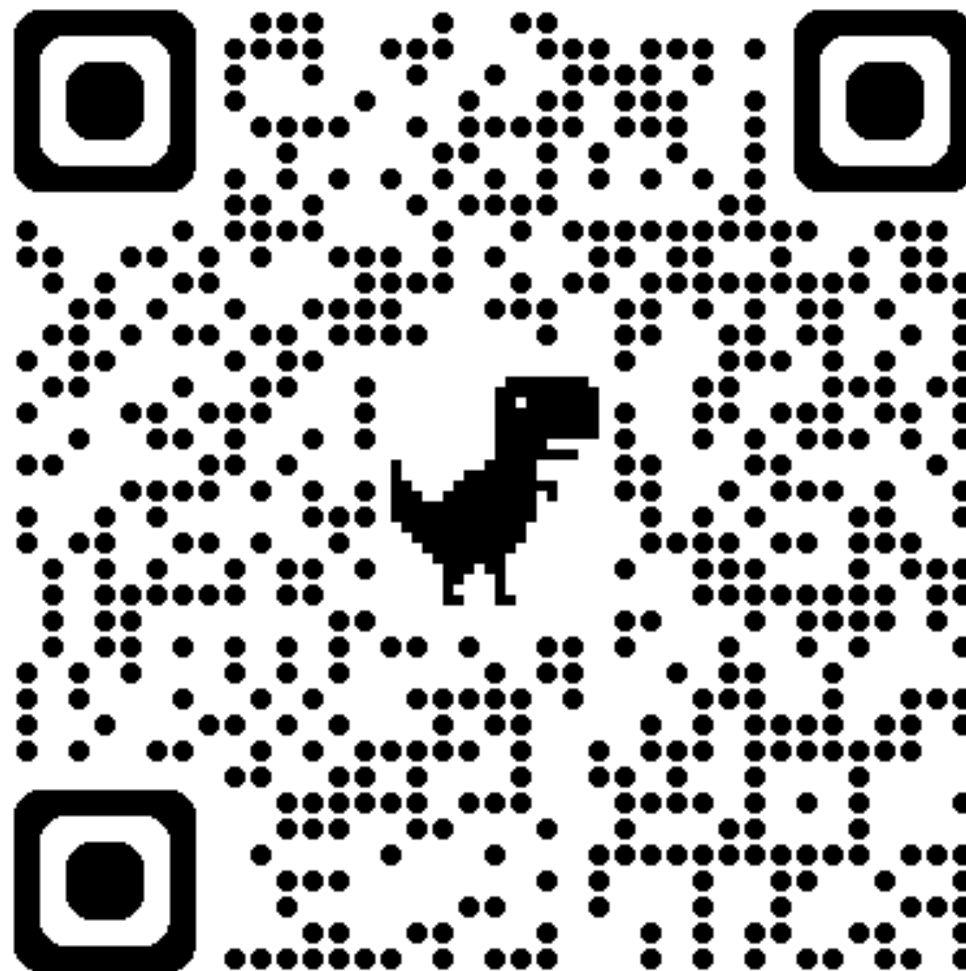


講義/課程資  
源下載



# 以AI之盾，防AI之矛

## 語音偵偽技術及應用工具簡介

中華電信研究院 前瞻所

2024/10/04



Refresh your life

# 大綱

- ◆課程資源
- ◆語音偵偽系統架構及評分重點
- ◆語音偵偽技術--ASVSpooF 競賽簡介
- ◆語音偵偽Baseline模型-- Wav2vec-AASIST
- ◆Mocking API介紹與使用方法
- ◆API整合AI模型 (加分)

# 課程資源

- ◆ 課程資源載點：[202410\\_Hackathon](#)
- ◆ Docker使用者：下載映像檔、API測試文本
- ◆ 無Docker替代方案：Anaconda
  - 下載anaconda：<https://docs.anaconda.com/anaconda/install/>
  - 下載專案程式碼 project\_spf\_v0\_api、環境配置文件 environment.yaml

## Installation

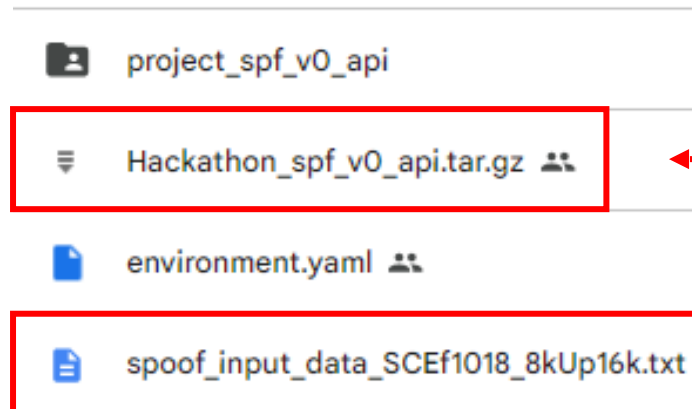
Review the system requirements listed below before installing Anaconda Distribution. If you don't want the hundreds of packages included with Anaconda, [install Miniconda](#), a mini version of Anaconda that includes just conda, its dependencies, and Python.

### System requirements

On Windows, macOS, and Linux, it is best to install Anaconda for the local user, which does not require administrator permissions and is the most robust type of installation. However, with administrator permissions, you can install Anaconda system wide.

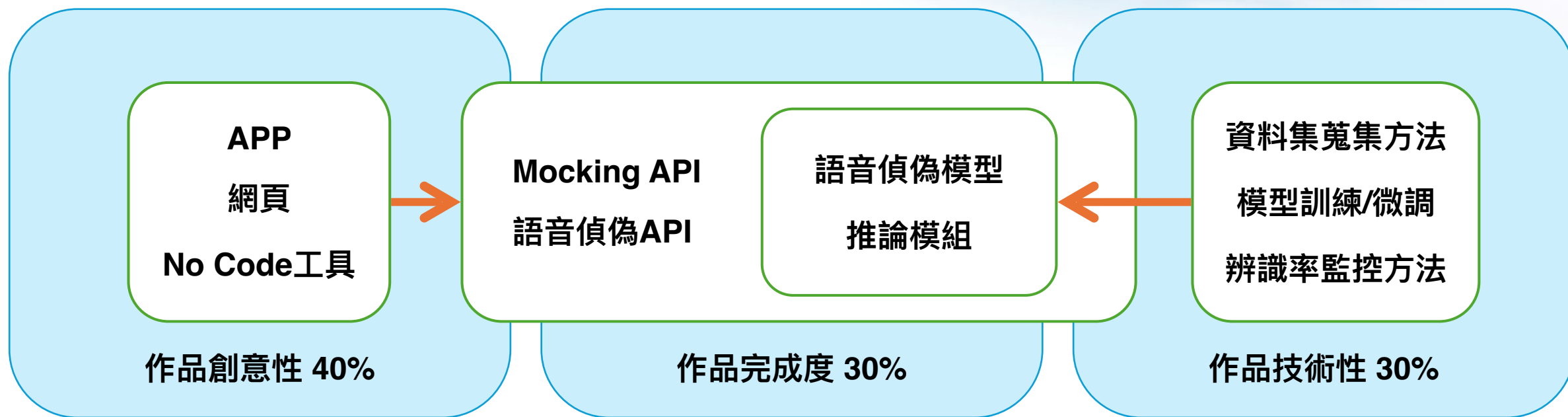
[Installing on Windows](#)  
[Installing on macOS](#)  
[Installing on Linux](#)

依電腦系統選擇載點



202410\_Hackathon 雲端資料夾

# 語音偵偽系統架構及評分重點

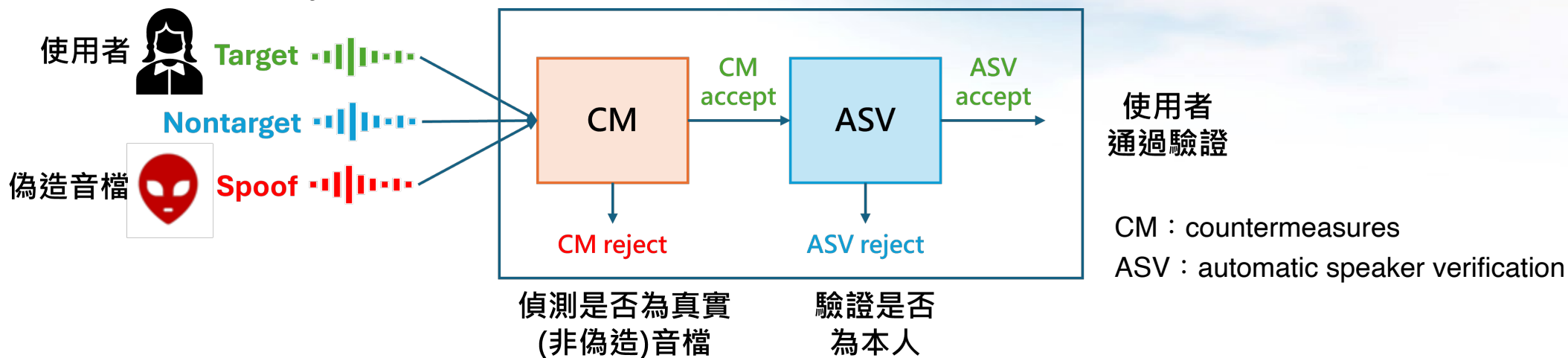


# 語音偵偽技術--ASVSpooof 競賽簡介

- ◆ 競賽目標: 檢測 TTS(text-to-speech)、VC(voice conversion)、語音回放、deepfake等欺騙語音攻擊

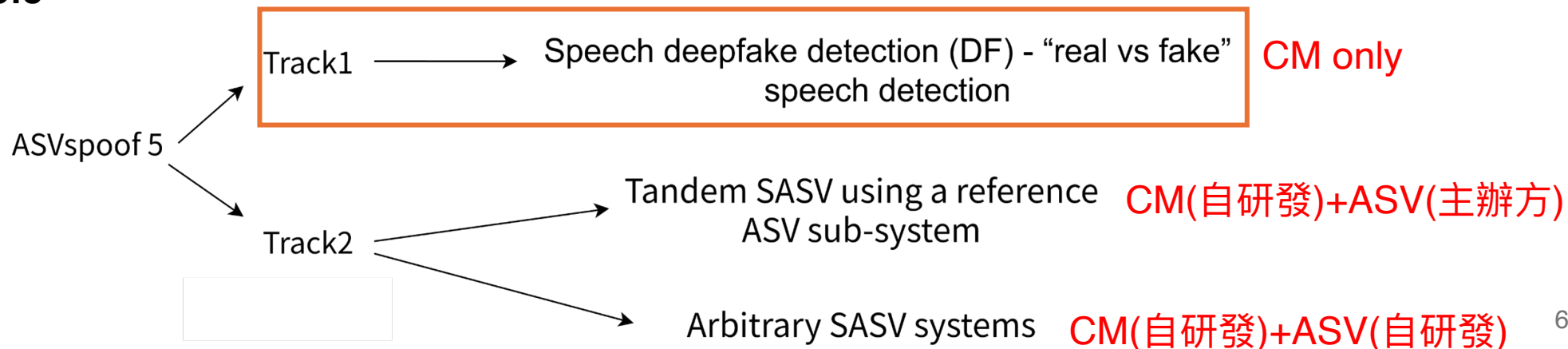
- ◆ **ASVspooof2021**

- LA task 包含 CM-only 與 CM+ASV



- ◆ **ASVspooof5**

- 任務



# 語音偵偽技術-- ASVSpooF 競賽資料集

## ◆ 競賽資訊

- <https://www.asvspoof.org/index2021.html> (ASVspoof2021)
- <https://www.asvspoof.org/> (ASVspoof5)

Spoof: 偽造

Bonafide: 真實

## 競賽資料集之使用歷程

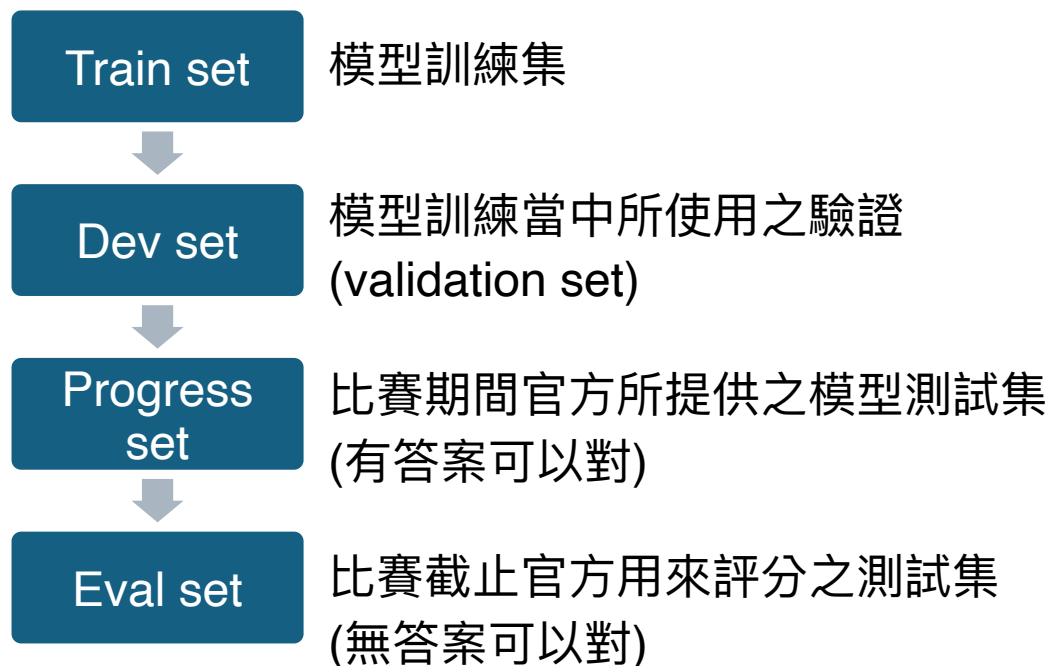


Table 2: Summary of the ASVspoof 5 challenge database on Track 1 we utilized.

| ID | Data-Set    | Subset   | utterances |         |
|----|-------------|----------|------------|---------|
|    |             |          | bona-fide  | spoofed |
| A  | ASVspoof 5  | Train    | 18,797     | 163,560 |
| B  |             | Dev      | 31,334     | 31,334  |
| C  | 2021 LA     | Progress | 1,676      | 14,788  |
| D  |             | Eval     | 14,816     | 133,360 |
| E  | 2019 LA     | Train    | 2,580      | 22,800  |
| F  |             | Dev      | 2,548      | 22,296  |
| G  |             | Eval     | 7,355      | 63,882  |
| H  | VCTK        | -        | 88,328     | -       |
| I  | In-the-Wild | -        | 19,963     | 11,816  |

## 訓練資料範例

```
19 T_1032 T_0000000018 F - A07 spoof
20 T_2386 T_0000000019 M - bonafide bonafide
21 T_3280 T_0000000020 M - A02 spoof
22 T_2498 T_0000000021 M - bonafide bonafide
```

(ID 音檔名稱 性別 - 合成方法 音檔類型)

# 語音偵偽技術-- ASVSpooF5 競賽結果

## ◆ Closed condition- 不可用預訓練模型

| Closed condition |     |        |        |           |       |           |     |        |        |           |       |
|------------------|-----|--------|--------|-----------|-------|-----------|-----|--------|--------|-----------|-------|
| #                | ID  | minDCF | actDCF | $C_{llr}$ | EER   | #         | ID  | minDCF | actDCF | $C_{llr}$ | EER   |
| • 1              | T32 | 0.244  | 0.996  | 0.946     | 8.61  | 準確率91.39% |     |        |        |           |       |
| • 2              | T47 | 0.266  | 0.338  | 0.609     | 9.18  |           |     |        |        |           |       |
| • 3              | T24 | 0.297  | 0.298  | 0.418     | 10.43 |           |     |        |        |           |       |
| • 4              | T45 | 0.395  | 1.000  | 0.852     | 14.33 |           |     |        |        |           |       |
| • 5              | T13 | 0.403  | 0.422  | 0.524     | 14.75 |           |     |        |        |           |       |
| 6                | -   | 0.408  | 0.430  | 0.551     | 14.16 | • 20      | T07 | 0.628  | 1.000  | 1.075     | 25.47 |
| 7                | -   | 0.439  | 0.633  | 0.853     | 17.09 | 21        | -   | 0.634  | 1.094  | 1.081     | 26.17 |
| • 8              | T46 | 0.478  | 1.000  | 1.051     | 20.45 | 22        | -   | 0.646  | 0.839  | 2.325     | 26.45 |
| • 9              | T23 | 0.531  | 1.000  | 1.117     | 20.13 | • 23      | T41 | 0.654  | 0.764  | 0.918     | 26.28 |
| 10               | -   | 0.534  | 1.000  | 1.023     | 19.10 | ◦ 24      | T06 | 0.660  | 1.000  | 1.116     | 28.41 |
| 11               | -   | 0.536  | 0.953  | 3.307     | 22.67 | 25        | -   |        |        |           |       |
| • 12             | T35 | 0.550  | 1.000  | 1.143     | 23.42 | 26        | -   |        |        |           |       |
| 13               | -   | 0.581  | 0.854  | 4.099     | 23.34 | 27        | -   |        |        |           |       |
| ◦ 14             | T48 | 0.581  | 0.935  | 3.192     | 23.63 | ◦ 28      | B02 |        |        |           |       |
| ◦ 15             | T19 | 0.589  | 0.688  | 1.328     | 24.59 | ◦ 29      | T44 |        |        |           |       |
| 16               | -   | 0.590  | 1.000  | 0.935     | 23.93 | 30        | -   |        |        |           |       |
|                  |     |        |        |           |       | ◦ 31      | B01 |        |        |           |       |

## ◆ Open condition- 可使用預訓練模型

| Open condition |     |        |        |           |      |           |     |        |        |           |       |
|----------------|-----|--------|--------|-----------|------|-----------|-----|--------|--------|-----------|-------|
| #              | ID  | minDCF | actDCF | $C_{llr}$ | EER  | #         | ID  | minDCF | actDCF | $C_{llr}$ | EER   |
| •▲ 1           | T45 | 0.075  | 1.000  | 0.792     | 2.59 | 準確率97.41% |     |        |        |           |       |
| •▲ 2           | T36 | 0.094  | 1.000  | 0.887     | 3.41 |           |     |        |        |           |       |
| 3              | -   | 0.094  | 0.138  | 0.193     | 3.42 |           |     |        |        |           |       |
| •▲ 4           | T23 | 0.112  | 1.000  | 0.918     | 4.16 |           |     |        |        |           |       |
| •▲ 5           | T43 | 0.115  | 0.573  | 0.956     | 4.04 |           |     |        |        |           |       |
| •▲ 6           | T13 | 0.130  | 0.141  | 0.379     | 4.50 | 18        | -   | 0.195  | 0.244  | 0.703     | 7.05  |
| •▲ 7           | T06 | 0.135  | 0.217  | 0.310     | 5.02 | •▲ 22     | T51 | 0.224  | 1.000  | 0.801     | 7.72  |
| 8              | -   | 0.141  | 0.529  | 0.615     | 4.89 | •▲ 23     | T46 | 0.225  | 1.000  | 1.031     | 9.36  |
| ◦▲ 9           | T31 | 0.150  | 0.224  | 0.556     | 5.56 | 24        | -   | 0.257  | 1.000  | 0.996     | 9.28  |
| •▲ 10          | T29 | 0.155  | 0.205  | 0.729     | 5.37 | 25        | -   | 0.264  | 0.704  | 2.189     | 10.32 |
| •▲ 11          | T35 | 0.161  | 1.000  | 1.038     | 5.93 | •△ 26     | T47 | 0.266  | 0.332  | 0.493     | 9.18  |
| 12             | -   | 0.167  | 0.167  | 0.235     | 5.77 | 27        | -   | 0.267  | 0.292  | 0.619     | 9.59  |
| •▲ 13          | T21 | 0.173  | 0.239  | 0.950     | 6.01 | •▲ 28     | T41 | 0.301  | 0.309  | 0.477     | 10.45 |
| ◦▲ 14          | T17 | 0.173  | 1.000  | 2.322     | 5.99 | 29        | -   | 0.412  | 0.427  | 0.719     | 14.25 |
| ◦▲ 15          | T19 | 0.174  | 0.309  | 0.476     | 6.06 | 30        | -   | 0.484  | 1.000  | 0.933     | 17.08 |
| 16             | -   | 0.184  | 1.000  | 0.876     | 6.35 | ◦△ 31     | T15 | 0.511  | 0.672  | 0.886     | 22.24 |
| 17             | -   | 0.193  | 1.000  | 0.834     | 6.67 | 32        | -   | 0.658  | 0.745  | 1.140     | 22.90 |
|                |     |        |        |           |      | 33        | -   | 0.797  | 1.000  | 0.992     | 35.72 |

(task: CM only)

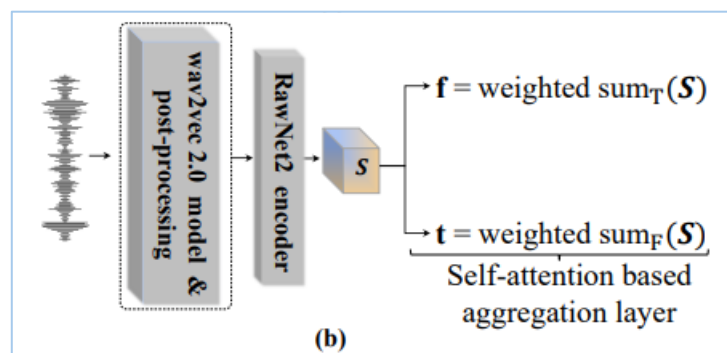
# 語音偵偽 Baseline 模型-- Wav2vec-AASIST

## ◆ 論文end-to-end框架wav2vec-AASIST\* (ASVspoo2021提出)

- Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation

論文連結: <https://arxiv.org/pdf/2202.12233>

## 2. The AASIST back-end



## 1. The wav2vec front-end

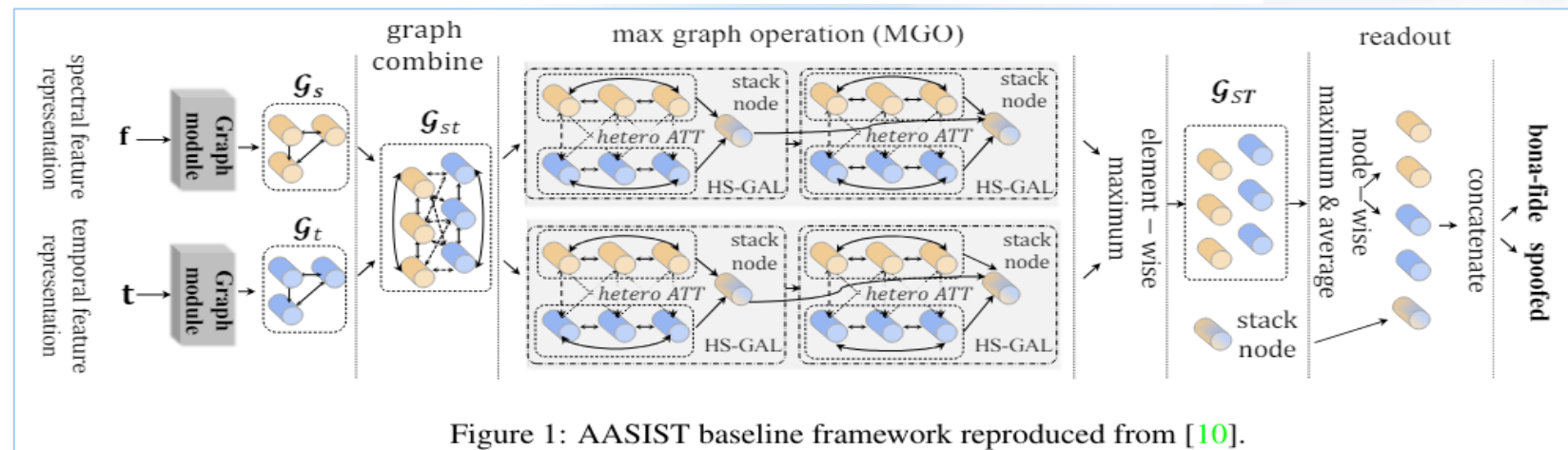


Figure 1: AASIST baseline framework reproduced from [10].

## ◆ 模型效能

### 於2021測試集

- Logical Access(LA)任務準確率達 99.18% (論文table2)
- DeepFake(DF)任務準確率達 97.15% (論文table3)

| front-end          | SA       | DA       | Pooled EER         | Pooled min t-DCF       |
|--------------------|----------|----------|--------------------|------------------------|
| sinc-layer         | ×        | ×        | 11.47 (11.95)      | 0.5081 (0.5139)        |
| wav2vec 2.0        | ×        | ×        | 6.15 (6.46)        | 0.3577 (0.3587)        |
| sinc-layer         | ✓        | ×        | 8.73 (11.61)       | 0.4285 (0.5203)        |
| wav2vec 2.0        | ✓        | ×        | 4.48 (6.15)        | 0.3094 (0.3482)        |
| sinc-layer         | ✓        | ✓        | 7.65 (7.87)        | 0.3894 (0.3960)        |
| <b>wav2vec 2.0</b> | <b>✓</b> | <b>✓</b> | <b>0.82 (1.00)</b> | <b>0.2066 (0.2120)</b> |

Table 2: Pooled EER and pooled min t-DCF results for the ASVspoo 2021 LA database evaluation set, for the sinc-layer and wav2vec 2.0 front-ends. SA refers to the self-attentive aggregation layer whereas DA refers to data augmentation. Results are the best (average) obtained from three runs of each experiment with different random seeds.

數值表示 best(average)%

# Wav2vec-AASIST程式碼、資料集與預訓練模型載點

- ◆ Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation 論文原始碼

- [https://github.com/TakHemlata/SSL\\_Anti-spoofing](https://github.com/TakHemlata/SSL_Anti-spoofing)
- 可找到模型檔供下載

## Pre-trained wav2vec 2.0 XLSR (300M)

Download the XLSR models from [here](#)

We also provide a pre-trained models. To use it you can run:

Pre-trained SSL antispoofing models are available for LA and DF [here](#)

```
CUDA_VISIBLE_DEVICES=0 python main_SSL_LA.py --track=LA --is_eval  
CUDA_VISIBLE_DEVICES=0 python main_SSL_DF.py --track=DF --is_eval
```

- 2019訓練資料載點: <https://datashare.ed.ac.uk/handle/10283/3336>
- ◆ 其他特徵擷取模型(wavLM)供參，可自行運用
  - <https://github.com/microsoft/unilm/tree/master/wavlm>

[http://\\${hostname}/spoofer\\_detector](http://${hostname}/spoofer_detector)

說明: 輸入欲辨識之音訊，輸出偽造語音之預測結果



Input

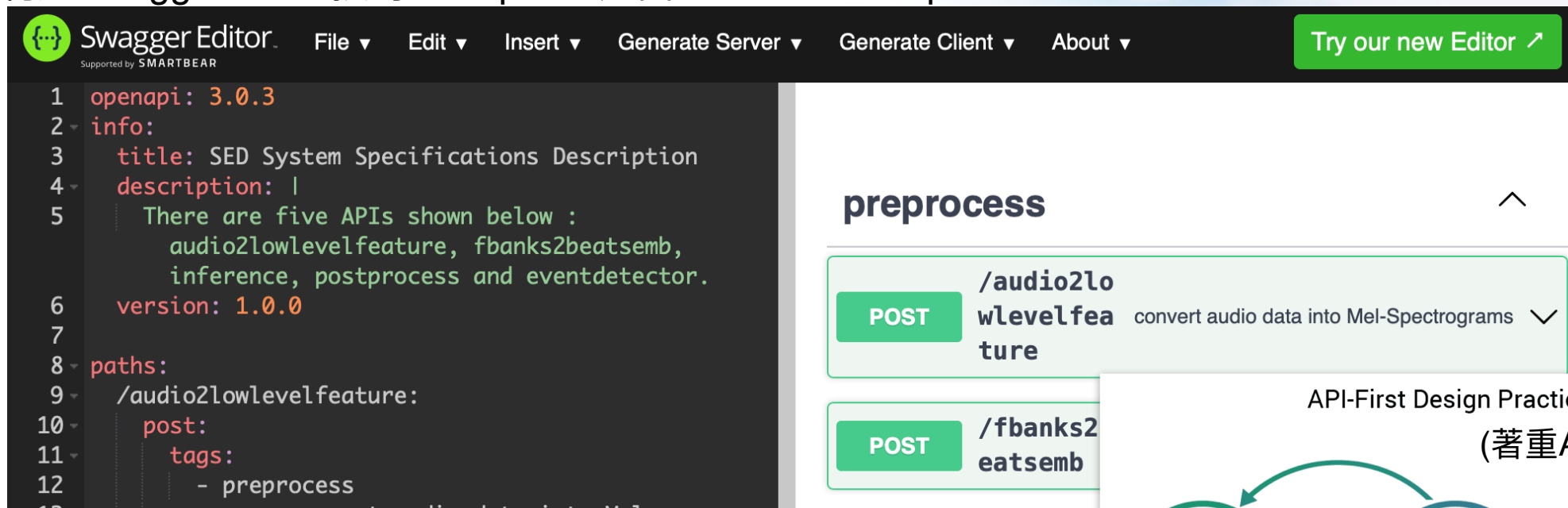
| 項次 | 欄位            | 欄位說明         | 必填 | 資料型態   | 備註                                          |
|----|---------------|--------------|----|--------|---------------------------------------------|
| 1  | reference_id  | reference ID | Y  | String |                                             |
| 2  | audio_data    | 辨識音檔         | Y  | String | 音檔格式：wav<br>format: binary (base64 encoded) |
| 3  | model_version | 模型版本         | Y  | String |                                             |
| 項次 | 欄位            | 欄位說明         | 必填 | 資料型態   | 備註                                          |

output

|     |                 |                       |   |         |                                                |
|-----|-----------------|-----------------------|---|---------|------------------------------------------------|
| 1   | reference_id    | reference ID          | Y | String  |                                                |
| 2   | uid             | unique transaction ID | Y | String  | 時間戳記-UUID(8)                                   |
| 3   | status_code     | 狀態碼                   | Y | Integer | 200：成功<br>500：伺服器端錯誤                           |
| 4   | processing_time | 程式執行時間                | Y | Object  |                                                |
| 4.1 | start_time      | 起始時間                  | Y | String  | "2021-08-27T15:29:59.668"<br>format: date-time |
| 4.2 | end_time        | 完成時間                  | Y | String  | "2021-08-27T15:29:59.699"<br>format: date-time |
| 4.3 | duration_in_s   | 持續時間(秒)               | Y | Number  | format: float                                  |
| 5   | error_message   | 音檔分析失敗原因              | N | String  | 音檔分析失敗時(status_code!=200)，回覆失敗原因               |
| 6   | result          | 事件種類的集合               | N | String  | real/fake                                      |
| 7   | confidence      | 單一事件信心分數              | N | Number  | format: float                                  |

# Mocking API 框架- Swagger

- ◆ Swagger：一種開源工具，有助於設計、創建、文件化API。其使用 OpenAPI 規範(以前稱swagger)格式來撰寫 API spec。
- ◆ 利用Swagger editor撰寫API spec並產出API server template



The screenshot shows the Swagger Editor interface. On the left, a code editor displays an OpenAPI specification snippet:

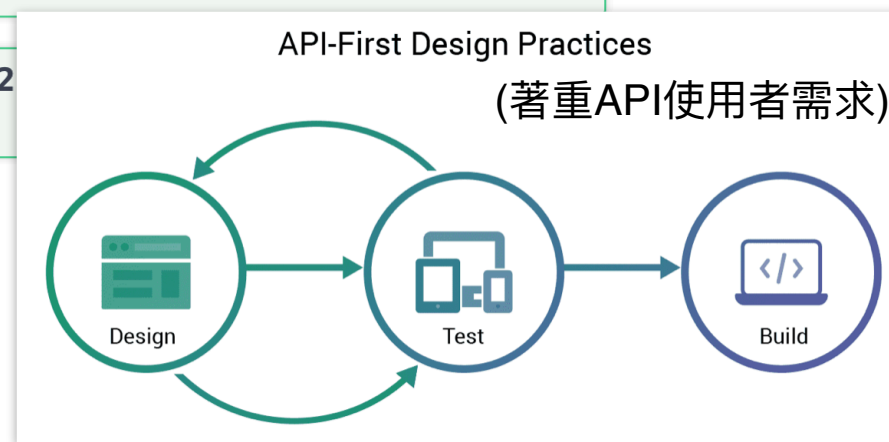
```
1 openapi: 3.0.3
2 info:
3   title: SED System Specifications Description
4   description: |
5     There are five APIs shown below :
6     audio2lowlevelfeature, fbanks2beatsemb,
7     inference, postprocess and eventdetector.
8   version: 1.0.0
9 paths:
10  /audio2lowlevelfeature:
11    post:
12      tags:
13        - preprocess
```

On the right, a panel titled "preprocess" lists two API endpoints:

- POST** `/audio2lowlevelfeature` convert audio data into Mel-Spectrograms
- POST** `/fbanks2beatsemb`

註： OpenAPI規範完整文件參照

<https://github.com/OAI/OpenAPI-Specification/blob/main/versions/3.0.3.md>



<https://softjourn.com/insights/5-benefits-of-api-first-design>

# 以Docker image部署Mocking API

- ◆ 下載映像檔 **Hackathon\_spf\_v0\_api.tar.gz**

註: 此環境python版本為 3.10.8

- ◆ Load image (範例環境: Ubuntu)

```
$docker load -i Hackathon_spf_v0_api.tar.gz
```

```
$docker images #確認上載成功
```

| REPOSITORY        | TAG | IMAGE ID    | CREATED    | SIZE |
|-------------------|-----|-------------|------------|------|
| hackathon_spf_api | v0  | fb8ef0a7275 | 3 days ago | 10GB |

- ◆ 啟動container (成功進入後會在 /home/docker/spoof\_docker 目錄下)

```
$ docker run --rm -it --name spf_api -p 8085:8085 -v /{your_path}/project_spf_v0_api:/home/docker/spoof_docker hackathon_spf_api:v0
```

視需求掛載資料夾

- ◆ 啟動API server

```
$ uvicorn "swagger_server. main :main" --host 0.0.0.0 --port 8085
```

```
root@9db1c8f72864:/home/docker/spoof_docker# uvicorn "swagger_server.__main__:main" --host 0.0.0.0 --port 8085
WARNING: ASGI app factory detected. Using it, but please consider setting the --factory flag explicitly.
INFO: Started server process [17]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8085 (Press CTRL+C to quit)
```

註：使用mac者(M1/M2晶片)若遇到錯誤 “The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested”

請改用指令 `docker run --platform linux/amd64 --rm -it --name spf_api -p 8085:8085 hackathon_spf_api:v0`

# Mocking API 使用方式(docker)

## ◆ 在container外部使用curl呼叫API範例

```
$curl -X 'POST' \  
  'http://localhost:8085/spoof_detector' \  
  -H 'accept: application/json' \  
  -H 'Authorization: Bearer 456123' \  
  -H 'Content-Type: application/json' \  
  -d @/{your_path}/spoof input data SCEf1018 8kUp16k.txt
```

### API之輸入資料

#### ➤ 小試身手:

1. 執行左邊指令查看回傳結果
2. 將 “model\_version”: “v0”  
改為 “model\_version”: “xxx” ，查看回傳結果

### 回傳結果(成功)

```
{  
  "confidence": 0.85,  
  "processing_time": {  
    "duration_in_s": 0.00021696090698242188,  
    "end_time": "2024-09-27T12:02:23.930",  
    "start_time": "2024-09-27T12:02:23.930"  
  },  
  "reference_id": "REF00000001",  
  "result": "real",  
  "status_code": 200,  
  "uid": "20210827142959-42f091f4"  
}
```

### 回傳結果(失敗)

```
{  
  "error_message": "incorrect input values",  
  "processing_time": {  
    "duration_in_s": 0.0007293224334716797,  
    "end_time": "2024-09-27T12:09:45.002",  
    "start_time": "2024-09-27T12:09:45.001"  
  },  
  "reference_id": "REF00000001",  
  "status_code": 400,  
  "uid": "20210827142959-42f091f4"  
}
```

## API server端log顯示

```
2024-09-27 12:02:23,930 - spoof-api - INFO - analyzing...  
2024-09-27 12:02:23,930 - spoof-api - INFO - {'reference_id': 'REF00000001', 'uid': '20210827142959-42f091f4', 'status_code': 200, 'processing_time': {'start_time': '2024-09-27T12:02:23.930', 'end_time': '2024-09-27T12:02:23.930', 'duration_in_s': 0.00021696090698242188}, 'result': 'real', 'confidence': 0.85}  
INFO: 172.17.0.1:44724 - "POST /spoof_detector HTTP/1.1" 200 OK
```

# Mocking API 使用方式(anaconda)

◆ 下載專案連結：[202410\\_Hackathon](#)

◆ 打開 anaconda prompt

```
$conda create -n hackathon_env python=3.10    #建立虛擬環境
```

```
$conda env list    #虛擬環境列表
```

```
$conda activate hackathon_env    #進入虛擬環境
```

```
$conda env update -n hackathon_env --file environment.yaml    # 更新環境(安裝所需套件)
```

```
$cd Downloads/project_spf_v0_api    #切到專案目錄下
```

```
$uvicorn "swagger_server.__main__:main" --host 0.0.0.0 --port 8085    #啟動API server
```

◆ 另開一個 anaconda prompt 視窗

(確認在目錄 /home/docker/spoof\_docker 下)

```
$python swagger_server/test/e2e_test.py    #執行測試腳本
```

◆ 於UI顯示API spec: <http://localhost:8085/ui/>

## Spoof Detect System Specifications Description <sup>1.0.0</sup> <sup>OAS3</sup>

/openapi.json

The main API is called spoof\_detector. You can input an audio data to detect its authenticity.

Servers  
▼

### security

GET /auth/{user\_id} Return JWT token

### analyze\_tool

POST /spoof\_detector detect events from audio data

# API 整合核心模型（加分）

## ◆ API開發相關程式

| 程式項目       | 檔案路徑                                                    |
|------------|---------------------------------------------------------|
| API spec定義 | ./swagger_server/swagger/swagger.yaml                   |
| API主程式     | ./swagger_server/controllers/analyze_tool_controller.py |
| API測試      | ./swagger_server/test/e2e_test.py                       |

/home/henacheng/sr\_2409\_Hackathon\_spf\_v0\_api/swagger\_server/

| Name         | Size (KB) |
|--------------|-----------|
| ..           |           |
| test         |           |
| log          |           |
| controllers  |           |
| __pycache__  |           |
| core         |           |
| models       |           |
| swagger      |           |
| __main__.py  | 1         |
| type_util.py | 1         |
| util.py      | 3         |
| __init__.py  | 0         |
| encoder.py   | 1         |

## ◆ 核心模型相關程式

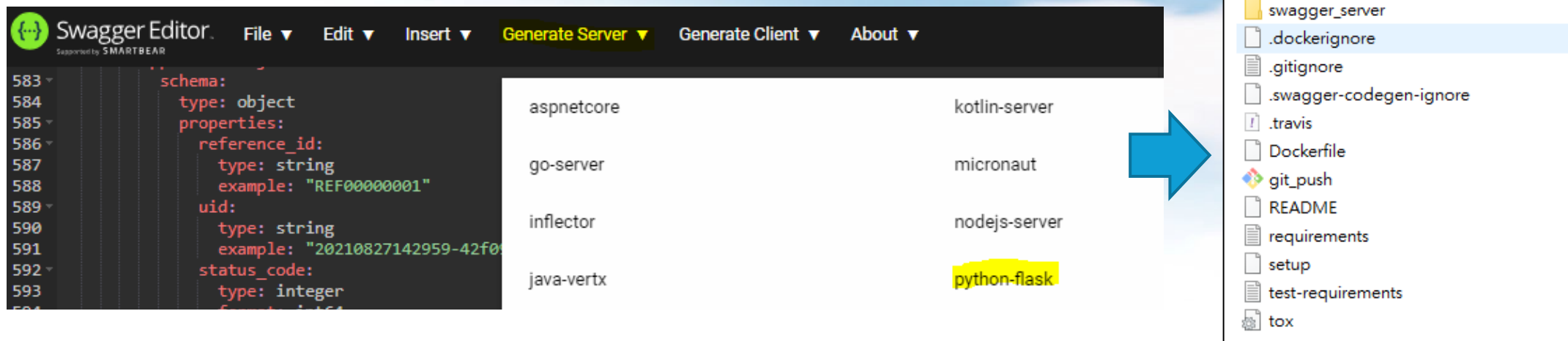
- baseline模型: github提供之模型參數檔
- 參考腳本[https://github.com/TakHemlata/SSL\\_Anti-spoofing/blob/main/model.py](https://github.com/TakHemlata/SSL_Anti-spoofing/blob/main/model.py)



# 附錄

# Swagger 工具使用- Swagger UI

## ① 於 swagger editor 生成 swagger server 的程式 template



## ② 按 README 安裝環境並啟動 swagger server

```
(test_swagger) C:\Users\henacheng\Downloads\python-flask-server-generated (2)>python -m swagger_server
* Serving Flask app '__main__' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: on
* Restarting with stat
* Debugger is active!
* Debugger PIN: 224-862-230
* Running on all addresses.
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://10.144.89.68:8080/ (Press CTRL+C to quit)
```

## ③ 打開網頁- Swagger UI (<http://localhost:8080/api/ui/>)

